MA3236 NONLINEAR PROGRAMMING
Semester 1, AY 2026/2027
Tutorial 8

Review: Lecture 16 — Lagrange–Newton and SQP

From KKT to a Newton step. For $\min f(\mathbf{x})$ subject to $\mathbf{g}(\mathbf{x}) = \mathbf{0}$ with C^2 data, set $L = f + \boldsymbol{\lambda}^T\mathbf{g}$, $\mathbf{A} = [\nabla g_1, \ldots, \nabla g_m]$ and $\mathbf{H}_L = \nabla^2_{\mathbf{xx}} L$. With $\mathbf{z} = (\mathbf{x}, \boldsymbol{\lambda})$, linearize the KKT system $F(\mathbf{z}) = \mathbf{0}$ to obtain the Newton step

$$F(\mathbf{z}) = \begin{pmatrix} \nabla_{\mathbf{x}} L \\ \mathbf{g} \end{pmatrix}, \qquad \underbrace{\begin{pmatrix} \mathbf{H}_L & \mathbf{A} \\ \mathbf{A}^T & 0 \end{pmatrix}}_{K(\mathbf{z})} \begin{pmatrix} \mathbf{d} \\ \boldsymbol{\delta} \end{pmatrix} = -F(\mathbf{z}), \qquad \mathbf{z}^+ = \mathbf{z} + (\mathbf{d}, \boldsymbol{\delta}).$$

- LICQ (rank $\mathbf{A} = m$) and $\mathbf{d}^T\mathbf{H}_L\mathbf{d} > 0$ for all $\mathbf{0} \neq \mathbf{d} \in \mathrm{Null}(\mathbf{A}^T)$ make K nonsingular; positive definiteness on all of $\mathbf{R}^n$ is unnecessary.
- At a KKT pair $\mathbf{z}^*$, nonsingular $K(\mathbf{z}^*)$ and locally Lipschitz K give quadratic convergence from a sufficiently close start: $\|\mathbf{z}^{(k+1)} - \mathbf{z}^*\| \leq C\|\mathbf{z}^{(k)} - \mathbf{z}^*\|^2$.

The same local step as an optimization problem. Sequential quadratic programming (SQP) minimizes a quadratic model over linearized constraints:

$$\min_{\mathbf{d}} \ \nabla f^T\mathbf{d} + \tfrac{1}{2}\mathbf{d}^T\mathbf{B}\mathbf{d} \quad \text{s.t.} \quad \mathbf{g} + \mathbf{A}^T\mathbf{d} = \mathbf{0} \qquad (\text{multiplier } \widehat{\boldsymbol{\lambda}}).$$

- Under LICQ and the positive reduced curvature above, $\mathbf{B} = \mathbf{H}_L$ gives $\mathbf{d}_{\mathrm{SQP}} = \mathbf{d}_{\mathrm{LN}}$ and $\widehat{\boldsymbol{\lambda}} = \boldsymbol{\lambda} + \boldsymbol{\delta}$ by the QP's KKT equations. Thus $\widehat{\boldsymbol{\lambda}}$ is the new multiplier estimate, not an increment.
- For $\mathbf{h} \leq \mathbf{0}$, also impose $\mathbf{h} + \mathbf{C}^T\mathbf{d} \leq \mathbf{0}$, where $\mathbf{C} = [\nabla h_1, \ldots, \nabla h_p]$; the QP chooses its own active set and multipliers $\widehat{\boldsymbol{\nu}} \geq \mathbf{0}$.
- If the Hessian lacks positive reduced curvature, approximate or modify it; if the linearized constraints are inconsistent, use restoration or relaxation.

Choose a step that makes actual progress. The equation $\mathbf{g} + \mathbf{A}^T\mathbf{d} = \mathbf{0}$ need not give $\mathbf{g}(\mathbf{x} + \mathbf{d}) = \mathbf{0}$. For equality SQP, combine objective and constraint violation in

$$\phi_\rho(\mathbf{x}) = f(\mathbf{x}) + \rho\|\mathbf{g}(\mathbf{x})\|_1, \qquad \rho > 0.$$

If $\mathbf{B} \succ 0$, $\mathbf{d} \neq \mathbf{0}$ and $\rho > \|\widehat{\boldsymbol{\lambda}}\|_\infty$, then $D\phi_\rho(\mathbf{x}; \mathbf{d}) < 0$. Backtracking selects $\alpha > 0$ satisfying

$$\phi_\rho(\mathbf{x} + \alpha\mathbf{d}) \leq \phi_\rho(\mathbf{x}) + \sigma\alpha D\phi_\rho(\mathbf{x}; \mathbf{d}), \qquad 0 < \sigma < 1.$$

Trust regions and filters are alternative globalization strategies. Merit decrease measures progress, but does not certify KKT conditions.

Check the original optimality conditions before stopping. For $\mathbf{g} = \mathbf{0}$, $\mathbf{h} \leq \mathbf{0}$, use the infinity-norm residuals

$$r_{\mathrm{feas}} = \max\{\|\mathbf{g}\|_\infty, \|[\mathbf{h}]_+\|_\infty\}, \qquad r_{\mathrm{stat}} = \|\nabla f + \mathbf{A}\boldsymbol{\lambda} + \mathbf{C}\boldsymbol{\nu}\|_\infty,$$
$$r_{\mathrm{dual}} = \|[-\boldsymbol{\nu}]_+\|_\infty, \qquad r_{\mathrm{comp}} = \|\boldsymbol{\nu} \odot \mathbf{h}\|_\infty.$$

Require every applicable residual to be at most $\varepsilon_{\mathrm{out}}$; a small step alone is insufficient.

Review: Lecture 17 — Barrier and primal interior point

Keep inequality iterates strictly inside. For $\mathbf{h}(\mathbf{x}) \leq \mathbf{0}$ and $\mu > 0$, retain equalities explicitly and solve

$$\min_{\mathbf{x}} \ \Phi_\mu(\mathbf{x}) = f(\mathbf{x}) - \mu \sum_{j=1}^{p} \log[-h_j(\mathbf{x})] \quad \text{s.t.} \quad \mathbf{g}(\mathbf{x}) = \mathbf{0}, \qquad \mathbf{h}(\mathbf{x}) < \mathbf{0}.$$

When all h_j are convex, convex (strictly convex) f makes Φ_μ convex (strictly convex) on its domain.

Read barrier stationarity as perturbed KKT. At an interior point, define $\nu_j = \mu/[-h_j] > 0$. Then

$$\begin{aligned} \nabla \Phi_\mu &= \nabla f + \mathbf{C}\boldsymbol{\nu}, \\ \nabla^2 \Phi_\mu &= \nabla^2 f + \sum_j \nu_j \nabla^2 h_j + \sum_j \frac{\mu}{h_j^2} \nabla h_j \nabla h_j^T. \end{aligned}$$

Thus an exact barrier stationary pair $(\mathbf{x}, \boldsymbol{\lambda})$ satisfies

$$\nabla f + \mathbf{A}\boldsymbol{\lambda} + \mathbf{C}\boldsymbol{\nu} = \mathbf{0}, \quad \mathbf{g} = \mathbf{0}, \quad \mathbf{h} < \mathbf{0}, \quad \boldsymbol{\nu} > \mathbf{0}, \quad -\nu_j h_j = \mu.$$

- Along this central path, $r_{\text{comp}} = \mu$ and $-\boldsymbol{\nu}^T \mathbf{h} = p\mu$. With C^1 data, $\mu^{(k)} \downarrow 0$, $\mathbf{x}^{(k)} \to \mathbf{x}^*$ and LICQ at $\mathbf{x}^*$, the limit satisfies KKT. For nonconvex problems, minimality needs a separate check.
- If an active multiplier tends to $\nu_j^* > 0$, then $-h_j = \mu/\nu_j = \Theta(\mu)$ and $\mu/h_j^2 = \Theta(1/\mu)$. The Hessian formula therefore explains the large normal curvature near active constraints.

Solve each barrier subproblem without leaving its domain. With equalities, use the L16 Newton system with $f \leftarrow \Phi_\mu$; without them, solve $\nabla^2 \Phi_\mu \, \mathbf{d} = -\nabla \Phi_\mu$.

- For affine $h_j = \mathbf{a}_j^T \mathbf{x} - b_j$, restrict the trial step to

  $$\alpha_{\max} = \min \left\{ 1, \ \tau \min_{\mathbf{a}_j^T \mathbf{d} > 0} \frac{-h_j(\mathbf{x})}{\mathbf{a}_j^T \mathbf{d}} \right\}, \qquad 0 < \tau < 1,$$

  which preserves $\mathbf{h} < \mathbf{0}$ (empty minimum: $+\infty$). For nonlinear $\mathbf{h}$, check actual $\mathbf{h}(\mathbf{x} + \alpha \mathbf{d}) < \mathbf{0}$ before logs.
- For a descent direction without equalities, backtrack from the feasible trial step until

  $$\Phi_\mu(\mathbf{x} + \alpha \mathbf{d}) \leq \Phi_\mu(\mathbf{x}) + \sigma \alpha \nabla \Phi_\mu(\mathbf{x})^T \mathbf{d}, \qquad 0 < \sigma < 1.$$

  With equalities, use an equality merit function in the line search.

Connect inner solves to the original problem.

- At outer iteration k, warm start the inner solve to $\varepsilon_{\text{in}}^{(k)}$, infer $\nu_j = \mu/[-h_j]$, and test the original KKT residuals against ε_{out}. Otherwise reduce μ and coordinate $\varepsilon_{\text{in}}^{(k)}$; a small inner gradient or small μ alone is insufficient.
- To obtain a strict start, use Phase I: $\min_{\mathbf{x},t}\{t : \mathbf{g}(\mathbf{x}) = \mathbf{0}, \ h_j(\mathbf{x}) \leq t\}$. A feasible $t < 0$ supplies one; an empty strict interior precludes it.

1. **One Lagrange–Newton step and its SQP interpretation.** Consider

$$\min_{\mathbf{x}\in\mathbf{R}^2} f(\mathbf{x}) = (x_1-1)^2 + (x_2-1)^2 \quad \text{s.t.} \quad g(\mathbf{x}) = x_1 + x_2 + x_1^2 = 0.$$

Start at $\mathbf{x}^{(0)} = (1,-2)^T$ and $\lambda^{(0)} = 0$.

(a) Form and solve the Lagrange–Newton system. Find the full-step primal and multiplier updates.

(b) Write and solve the equality SQP subproblem with $\mathbf{B}^{(0)} = \mathbf{H}_L^{(0)}$. Verify its multiplier's relation to the Newton multiplier update, and check both linearized and actual feasibility of the new point.

Solution. (a) Differentiation gives

$$\nabla f = \begin{pmatrix} 2x_1 - 2 \\ 2x_2 - 2 \end{pmatrix}, \qquad \nabla g = \begin{pmatrix} 1 + 2x_1 \\ 1 \end{pmatrix}, \qquad \mathbf{H}_L = \begin{pmatrix} 2 + 2\lambda & 0 \\ 0 & 2 \end{pmatrix}.$$

At the initial pair, $g = 0$ and $\nabla_{\mathbf{X}} L = (0,-6)^T$. The Newton equations are therefore

$$\begin{pmatrix} 2 & 0 & 3 \\ 0 & 2 & 1 \\ 3 & 1 & 0 \end{pmatrix} \begin{pmatrix} d_1^{(0)} \\ d_2^{(0)} \\ \delta^{(0)} \end{pmatrix} = \begin{pmatrix} 0 \\ 6 \\ 0 \end{pmatrix}.$$

The last equation gives $d_2^{(0)} = -3d_1^{(0)}$, and the first gives $\delta^{(0)} = -2d_1^{(0)}/3$. Substituting in the second yields $-(20/3)d_1^{(0)} = 6$. Hence

$$\mathbf{d}^{(0)} = (-9/10, 27/10)^T, \qquad \delta^{(0)} = 3/5,$$

$$\mathbf{x}^{(1)} = (1/10, 7/10)^T, \qquad \lambda^{(1)} = 3/5.$$

(b) The SQP model is

$$\min_{\mathbf{d}} \; -6d_2 + d_1^2 + d_2^2 \quad \text{s.t.} \quad 3d_1 + d_2 = 0.$$

Eliminating $d_2 = -3d_1$ gives $10d_1^2 + 18d_1$, whose unique minimizer is $d_1 = -9/10$. Its stationarity equations give $\widehat{\lambda} = 3/5 = \lambda^{(0)} + \delta^{(0)}$. Thus the two methods produce the same primal step. The linearized equality holds, but direct substitution gives

$$g(\mathbf{x}^{(1)}) = \frac{1}{10} + \frac{7}{10} + \frac{1}{100} = \frac{81}{100} \neq 0.$$

The omitted quadratic term $(d_1^{(0)})^2$ is precisely this feasibility error.

2. **A barrier path approaching an active constraint.** Let $\mathbf{a} = (1, 2)^T$ and consider

$$\min_{\mathbf{x}\in\mathbf{R}^2} f(\mathbf{x}) = \|\mathbf{x} - \mathbf{a}\|_2^2 \quad \text{s.t.} \quad h(\mathbf{x}) = \mathbf{a}^T\mathbf{x} - 1 \le 0.$$

(a) Form the logarithmic-barrier objective and state its domain. Find its unique minimizer $\mathbf{x}(\mu)$ for every $\mu > 0$.

(b) Compute the inferred multiplier $\nu(\mu)$ and verify $-\nu(\mu)h(\mathbf{x}(\mu)) = \mu$. Find the limiting primal point and multiplier as $\mu \downarrow 0$, and check their KKT conditions.

Solution. (a) The barrier is

$$\Phi_\mu(\mathbf{x}) = \|\mathbf{x} - \mathbf{a}\|_2^2 - \mu \log(1 - \mathbf{a}^T\mathbf{x}), \qquad \mathbf{a}^T\mathbf{x} < 1.$$

Its gradient and Hessian are

$$\nabla\Phi_\mu = 2(\mathbf{x} - \mathbf{a}) + \frac{\mu}{1 - \mathbf{a}^T\mathbf{x}}\mathbf{a}, \qquad \nabla^2\Phi_\mu = 2\mathbf{I} + \frac{\mu}{(1 - \mathbf{a}^T\mathbf{x})^2}\mathbf{a}\mathbf{a}^T.$$

The Hessian is positive definite throughout the convex domain. At a stationary point, the gradient equation forces $\mathbf{x} = t\mathbf{a}$. Since $\mathbf{a}^T\mathbf{a} = 5$, we need

$$2(t - 1) + \frac{\mu}{1 - 5t} = 0, \qquad t < \frac{1}{5}.$$

Multiplying by $1 - 5t$ gives $10t^2 - 12t + 2 - \mu = 0$. The two algebraic roots are $(6 \pm \sqrt{16 + 10\mu})/10$; only the minus sign satisfies the domain restriction. Therefore

$$\boxed{\mathbf{x}(\mu) = t(\mu)\mathbf{a}, \qquad t(\mu) = \frac{6 - \sqrt{16 + 10\mu}}{10}.}$$

This stationary point exists for every $\mu > 0$, and strict convexity makes it the unique global barrier minimizer.

(b) Put $D = \sqrt{16 + 10\mu}$. The slack and inferred multiplier are

$$s(\mu) = 1 - 5t(\mu) = \frac{D - 4}{2} > 0, \qquad \nu(\mu) = \frac{\mu}{s(\mu)} = \frac{D + 4}{5} = 2(1 - t(\mu)).$$

Their product is $(D^2 - 16)/10 = \mu$, as required. Taking limits gives

$$\mathbf{x}^* = (1/5, 2/5)^T, \qquad \nu^* = 8/5.$$

Here $h(\mathbf{x}^*) = 0$, $\nu^* > 0$, and

$$\nabla f(\mathbf{x}^*) + \nu^*\nabla h = 2(\mathbf{a}/5 - \mathbf{a}) + (8/5)\mathbf{a} = \mathbf{0}.$$

Primal feasibility, dual feasibility, stationarity, and complementarity all hold. Convex KKT sufficiency and strict convexity also show that $\mathbf{x}^*$ is the unique minimizer of the original problem.

3. **A feasible interior Newton step.** For the problem in Question 2, use $\mu = 1/10$ and $\mathbf{x}^{(0)} = \mathbf{0}$.

   (a) Evaluate $\nabla\Phi_\mu(\mathbf{x}^{(0)})$ and $\nabla^2\Phi_\mu(\mathbf{x}^{(0)})$, and compute the Newton direction $\mathbf{d}^{(0)}$. Does the full step stay in the barrier domain?

   (b) With $\tau = 9/10$, find the fraction-to-boundary trial length and point. At this point, infer ν and calculate r_{comp}. Is this point accurate enough for the outer method to stop with $\varepsilon_{\text{out}} = 10^{-3}$?

Solution. (a) Using the derivatives from Question 2,

$$\nabla\Phi_\mu(\mathbf{0}) = -\frac{19}{10}\mathbf{a}, \qquad \nabla^2\Phi_\mu(\mathbf{0}) = 2\mathbf{I} + \frac{1}{10}\mathbf{a}\mathbf{a}^T.$$

The Hessian sends $\mathbf{a}$ to $(2 + 5/10)\mathbf{a} = (5/2)\mathbf{a}$, so the Newton equation gives $\mathbf{d}^{(0)} = (19/25)\mathbf{a}$. The full step has $h(\mathbf{d}^{(0)}) = 19/5 - 1 = 14/5 > 0$ and leaves the barrier domain.

(b) The current slack is 1 and $\nabla h^T\mathbf{d}^{(0)} = 19/5$. Thus

$$\alpha_{\max} = \min\left\{1, \frac{9}{10}\frac{1}{19/5}\right\} = \frac{9}{38}, \qquad \mathbf{x}_{\text{trial}} = \frac{9}{50}\mathbf{a}.$$

Its slack is $1 - \mathbf{a}^T\mathbf{x}_{\text{trial}} = 1/10 > 0$, so the trial point stays inside the barrier domain. The inferred multiplier and complementarity residual are

$$\nu = \frac{\mu}{-h(\mathbf{x}_{\text{trial}})} = \frac{1/10}{1/10} = 1, \qquad r_{\text{comp}} = \nu[-h(\mathbf{x}_{\text{trial}})] = \frac{1}{10}.$$

Since $r_{\text{comp}} = 10^{-1} > 10^{-3}$, the point cannot satisfy the outer stopping test, regardless of its stationarity residual. The boundary safeguard only supplies a valid trial point; a line search is still needed before accepting an inner step.

4. **Merit decrease after a curved-constraint step (challenge).** Return to Question 1 and its direction $\mathbf{d}^{(0)}$. Use $\phi_\rho = f + \rho|g|$, with a fixed $\rho > 0$.

   (a) Find $\phi_\rho(\mathbf{x}^{(0)} + \alpha\mathbf{d}^{(0)}) - \phi_\rho(\mathbf{x}^{(0)})$ for $0 < \alpha \leq 1$. Which values of ρ make the full step strictly decrease the merit function?

   (b) Take $\rho = 20$ and a backtracking rule that tries $\alpha = 1, 1/2, 1/4, \ldots$ until the merit value strictly decreases. Find the first accepted step. Explain, using the orders of the objective change and constraint violation, why sufficiently small positive steps work for every fixed finite ρ, even when the full step fails.

Solution. (a) Along the line,

$$\mathbf{x}^{(0)} + \alpha\mathbf{d}^{(0)} = \begin{pmatrix} 1 - 9\alpha/10 \\ -2 + 27\alpha/10 \end{pmatrix}.$$

Substitution gives

$$f(\mathbf{x}^{(0)} + \alpha\mathbf{d}^{(0)}) = 9 - \frac{81}{5}\alpha + \frac{81}{10}\alpha^2, \qquad g(\mathbf{x}^{(0)} + \alpha\mathbf{d}^{(0)}) = \frac{81}{100}\alpha^2.$$

Since $g(\mathbf{x}^{(0)}) = 0$ and the new constraint value is nonnegative,

$$\phi_\rho(\mathbf{x}^{(0)} + \alpha\mathbf{d}^{(0)}) - \phi_\rho(\mathbf{x}^{(0)}) = \frac{81}{100}\alpha\big[(10 + \rho)\alpha - 20\big].$$

At $\alpha = 1$, this equals $81(\rho - 10)/100$. Hence the full step strictly decreases the merit value exactly when $0 < \rho < 10$.

(b) For $\rho = 20$, strict decrease holds when $0 < \alpha < 2/3$. The full step fails and $\alpha = 1/2$ succeeds; it is the first accepted step under the stated rule.

To see the mechanism, the objective's directional derivative is $\nabla f(\mathbf{x}^{(0)})^T\mathbf{d}^{(0)} = -81/5 < 0$, while $\nabla g(\mathbf{x}^{(0)})^T\mathbf{d}^{(0)} = 0$. Thus the objective decreases to first order in α, whereas the constraint violation enters only at second order:

$$\phi_\rho(\mathbf{x}^{(0)} + \alpha\mathbf{d}^{(0)}) - \phi_\rho(\mathbf{x}^{(0)}) = -\frac{81}{5}\alpha + O(\alpha^2).$$

For any fixed finite ρ, the negative linear term dominates as $\alpha \downarrow 0$. Increasing ρ can require shorter steps, but does not remove all acceptable steps in this example. This strict-decrease rule is the one specified in the question; an Armijo rule would require its additional sufficient-decrease margin.